INCH-POUND

MIL-T-52991A(ME)
18 August 1992
SUPERSEDING
MIL-T-52991(ME)
17 March 1981

MILITARY SPECIFICATION

TABLE, SCRIBING, TRACING, DRAFTING:

FLIP TOP

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a combination flip-top scribing, tracing, and drafting table for use in a mobile van application.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1898	- Cushioning Material, Packaging, (Cellulosic, Water Absorbent).
A-A-52301	- Drawing Board and Drawing Board Trestles.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-T-60	- Tape: Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6675

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MILITARY

MIL-P-116 - Preservation, Packaging, Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 512 - Tube, Metallic.
A 513 - Tube, Metallic.
A 519 - Tube, Metallic.
D 3951 - Standard Practice for Commercial Packaging.
D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association International, 60 Batterymarch Park, Quincy, MA 02269.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail

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specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The table shall be as shown in figures 1 and 2, and as specified herein. The table shall consist of a frame, drawing board-light box, top and drawer module. The drawer module shall swing clear of the pivoting work surfaces. The drawing board-light box top shall be a pivoting, flip-flop action type for ready conversion and security in either a work or mobile transport mode.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The table shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the table may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the table may be newly fabricated from recovered materials to the maximum extent practicable, provided the table produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the table.

3.5 Frame. The frame for the table shall be constructed of square steel tubing conforming to ASTM A 512, A 513 or A 519 with a square cross-sectional stock not less than 2 inches. Joints shall be welded or double bolted to assure stability and ready assembly. Overall dimensions shall be not more than 47 inches wide, approximately 34 inches deep, and approximately 42 inches high. Each leg shall have a locking level device independent of tie-downs required for mobile application.

3.5 Drawing board-light box top. The top shall be a combination drafting board-light box design with a back-to-back construction, pivot-point and trunion bracket on each side for a flip-flop pivoting action, and adjustable locking stops

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at fixed positions of an elevation to the horizontal. The elevations shall be approximately 5 and 10 degrees. The front edge of both the drawing board and light box shall be provided with drawing guards. The left and right edges shall have a metal strip to accommodate the use of a T-square.

3.5.1 Drawing board. The drawing board shall have a one-sided working surface and shall be approximately 42 inches wide, approximately 31 inches deep, and approximately 1 inch thick. The material and construction of the board shall be in accordance with A-A-52301.

3.5.2 Light box. The light box shall have a working surface not less than 30 inches long and 30 inches wide. The box shall be deep enough to house four 30-inch fluorescent fixtures, ballasts and lamps. Supports, lamps, fixtures, and reflecting surfaces shall be arranged for diffusion of uniform stable lighting. The glass plate shall be a safety glass, secured to the light box for flip-top action, removable for maintenance and cleaning, and mounted to prevent breakage in both the work and the transport mode, which is with the glass in the bottom positions.

3.5.2.1 Light diffusion. Light diffusion shall be accomplished by any of the following methods:

a. Surface of glass fine-grained.
b. Opal glass.
c. Opaline plastic sheet.

3.6 Drawer module. The table shall be fitted with a two drawer module that will swing forward out of the path of the flip-top action of the table top, around the right front leg of the frame. The module shall be positioned by a spring-loaded clasp remotely released by a lever. The release lever shall be located on the upper right portion of the drawer module. The clasp action shall securely fasten and position the module to the side frame in a 3-point control with guide plates or rests. The drawer module shall be provided with a mechanism to secure the drawers when the drawers are closed. The drawer module shall be approximately 12 inches wide by approximately 26 inches deep by approximately 18 inches high and shall be an accessory to, but not a part of the table frame. The module shall also be lockable with a single key. The drawers shall be provided with handles suitable for a one hand grasp.

3.7 Tray. When specified (see 6.2), the tray shall be approximately 12 inches long by 7 inches wide. The height shall be a diagonal approximately 3 inches to 1-1/2 inch. The method of temporarily securing the tray to the drawer module shall be by magnets and hooks (see figure 2).

3.8 Electrical system. The electrical system shall operate from 120-volt, 60 Hertz (Hz), single phase current. All electrical equipment and wiring shall conform to NFPA 70.

3.8.1 Power cord, switches, and receptacles. Two electrical receptacles, a fuse and an on-off switch shall be mounted vertically on the top part of the leg. The power input cord shall be approximately 9 feet long. The switch shall be "on" in the up position and "off" in the down position. The positions of the on-off

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switch shall be labeled to indicate "on" and "off". The electrical outlets shall be labeled "120 volt, 60 Hz, 1 PHASE". The electrical current rating of the fuse shall be permanently marked adjacent to the fuse holder.

3.9 Human factors engineering. The table shall comply with the design criteria of MIL-STD-1472. Special design emphasis shall be given, but not limited to, MIL-STD-1472, general requirements (4), controls (5.4), labeling (5.5), workspace design (5.7), design for maintainer (5.9), and hazards and safety (5.13), as applicable.

3.10 Identification marking. Each table shall be identified in accordance with MIL-STD-130.

3.11 Treatment and painting. The table shall be cleaned, treated with a primer and painted in accordance with accepted industry practice. Unless otherwise specified (see 6.2), the color shall be light green.

3.12 Workmanship. All parts, components, and assemblies of the table including castings, forgings, molded parts, stampings, bearings, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Nonfunctional sharp edges, projecting points and welding burrs shall be avoided. The glass working surface shall not be cracked, chipped, or broken. Drawers shall not bind and shall fit the drawer opening. Glue joints shall not have any looseness of wood pieces, wide gaps in seams, lack of rigidity or presence of warp. The paints shall not chip, streak, or flake.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First produced table (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The table shall be examined as specified herein for the following defects:

101. Materials not as specified (see 3.3).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.3).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the table (see 3.3.4).
106. Frame not as specified (see 3.4).
107. Drawing board-light box top not as specified (see 3.5).
108. Drawer module not as specified (see 3.6).
109. Tray not as specified (see 3.7).
110. Electrical system not as specified (see 3.8).
111. Human factors engineering not as specified (see 3.9).
112. Identification not as specified (see 3.10).
113. Treating and painting not as specified (see 3.11).
114. Workmanship not as specified (see 3.12).

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4.5.2 Tests.

4.5.2.1 Pivoting (flip flop) conversion. With the table assembled and the drawing board-light box in the drawing board mode, test the working surface for squared edges, stability and elevation positions. Unlatch and swing the drawer module forward around the right forward leg. Release the elevation fixture and rotate the drawing board-light box on its pivots and trunions to the light box mode. The conversion shall be made without binding and with clearance of both frame and drawer module. After the drawer module has been swung into working position, it shall be tested for stability in its mounts and spring latch by pushing and pulling on the module. Evidence that the working surface edges are not square, or evidence of binding during pivoting, rotating and elevating, as evidence of instability or lack of clearance shall constitute failure of this test (see 3.6).

4.6 Inspection of packaging.

4.6.1 First article inspection.

4.6.1.1 Examination. The first article shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

115.	Materials or containers not as specified.	Each incorrect material or container shall constitute one defect (see 5.3.1).
116.	Unpainted exterior metal surfaces requiring a contact preservative	not coated as specified (see 5.2.1.1.
117.	Lamps not secured as specified for level A (see 5.2.1.2).	
118.	Openings into electrical components not sealed as specified for	level A (see 5.2.1.2).
119.	Electrical cord not secured as specified for level A (see 5.2.1.2).	
120.	Table top and drawer section not secured as specified for level A (see	5.2.1.3).
121.	Glass working surface not cushioned as specified for level A (see	5.2.1.3).
122.	Table not covered with a shroud as specified for level A (see	5.2.1.4).
123.	Blocking, bracing and anchoring not as specified for level A or	level B (see 5.3.1 and 5.3.2).
124.	Strapping not as specified for level A or level B (see 5.3.1 and	5.3.2).

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125. Marking illegible, incorrect, incomplete or missing (see 5.4).

5. PACKAGING

5.1 First article pack. Unless otherwise specified (see 6.2), the contractor shall furnish a first article pack for examination within the time frame specified (see 6.2), to prove prior to starting production packaging that the applied preservation, packing and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The first article pack may be prepared utilizing either the first-produced table or a production table. When the first-produced table is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first-produced and production tables.

5.2 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Unpainted metal surfaces. Unpainted exterior metal surfaces of the table requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with MIL-P-116.

5.2.1.2 Electrical components. The fluorescent lamps shall be secured in their sockets with tape conforming to PPP-T-60, type IV. Openings into electrical components such as switches and outlets shall be sealed with tape specified herein. The cord shall be coiled and the coil secured with not less than three evenly spaced ties of cotton tape or twine.

5.2.1.3 Table top and drawer section. The table top and drawer section shall be secured. The glass working surface shall be covered with minimum 1-inch-thick cushioning material conforming to A-A-1898, grade II, class B, style and size optional. The cushioning material shall be secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.4 Complete table. The complete table, with components preserved as specified herein, shall be covered with a shroud fabricated from barrier material conforming to PPP-B-1055, class C-1 or C-2. The shroud shall be secured with tape conforming to PPP-T-60, type IV.

5.2.2 Commercial. Each table shall be preserved in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, level B, or Commercial, as specified (see 6.2).

5.3.1 Level A. Each complete table, preserved as specified in 5.2, shall be packed in a close-fitting box conforming to PPP-B-621, class 2, or PPP-B-601,

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overseas type. Blocking, bracing and anchoring of the table within the box shall be in accordance with MIL-STD-1186. Strapping shall conform to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.3.2 Level B. Packing of tables shall be as specified in 5.3.1 for level A, except that boxes shall be domestic class or type and strapping may be zinc-coated.

5.3.3 Commercial. Tables, preserved as specified in 5.2, shall be packed in accordance with ASTM D 3951.

5.4 Marking.

5.4.1 Marking, level A. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking shall be in accordance with ASTM D 3951. In addition, each shipping container shall display its gross weight and cube.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The table is used in the Cartographic Image Base Products, and Military Information Subsystems of the Mobile Topographic Support System (TSS).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of the first article (see 3.2).
d. When a tray is required (see 3.8).
e. Color required if other than as specified (see 3.12).
f. When a first article pack is not required (see 5.1).
g. Time frame required for submission of the first article pack, when applicable (see 5.1).
h. Degree of preservation and packing required (see 5.2 and 5.3).
i. Any special marking required (see 5.4).

6.3 First article. When a first article inspection is required, the item(s) should be an initial production model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence

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with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Information figures. Figures 1 and 2 show a type of table and tray which has been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of another table and tray which conforms to this specification.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Subject term (key-word) listing.

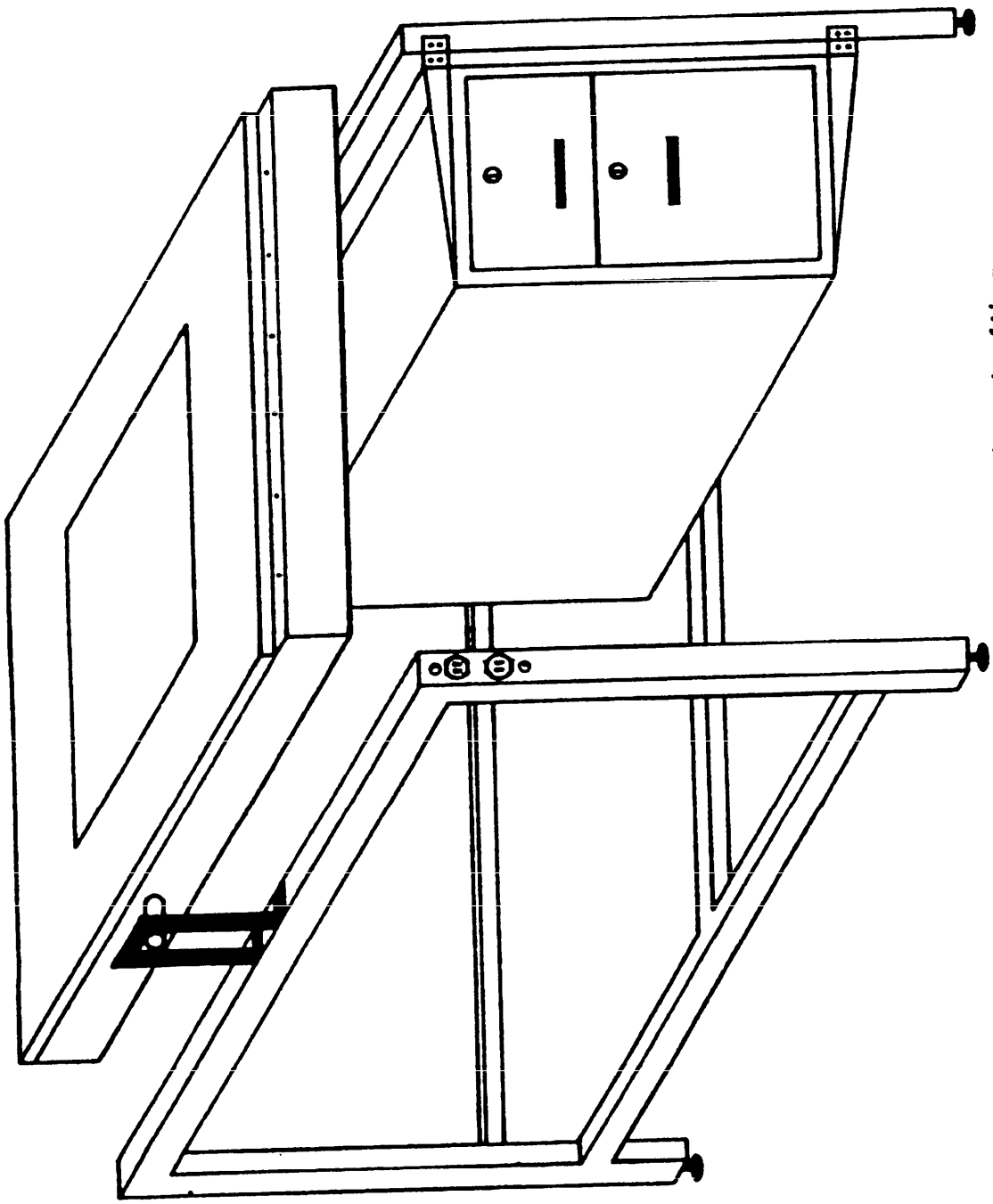
Counter
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Desk
Design

Custodians:
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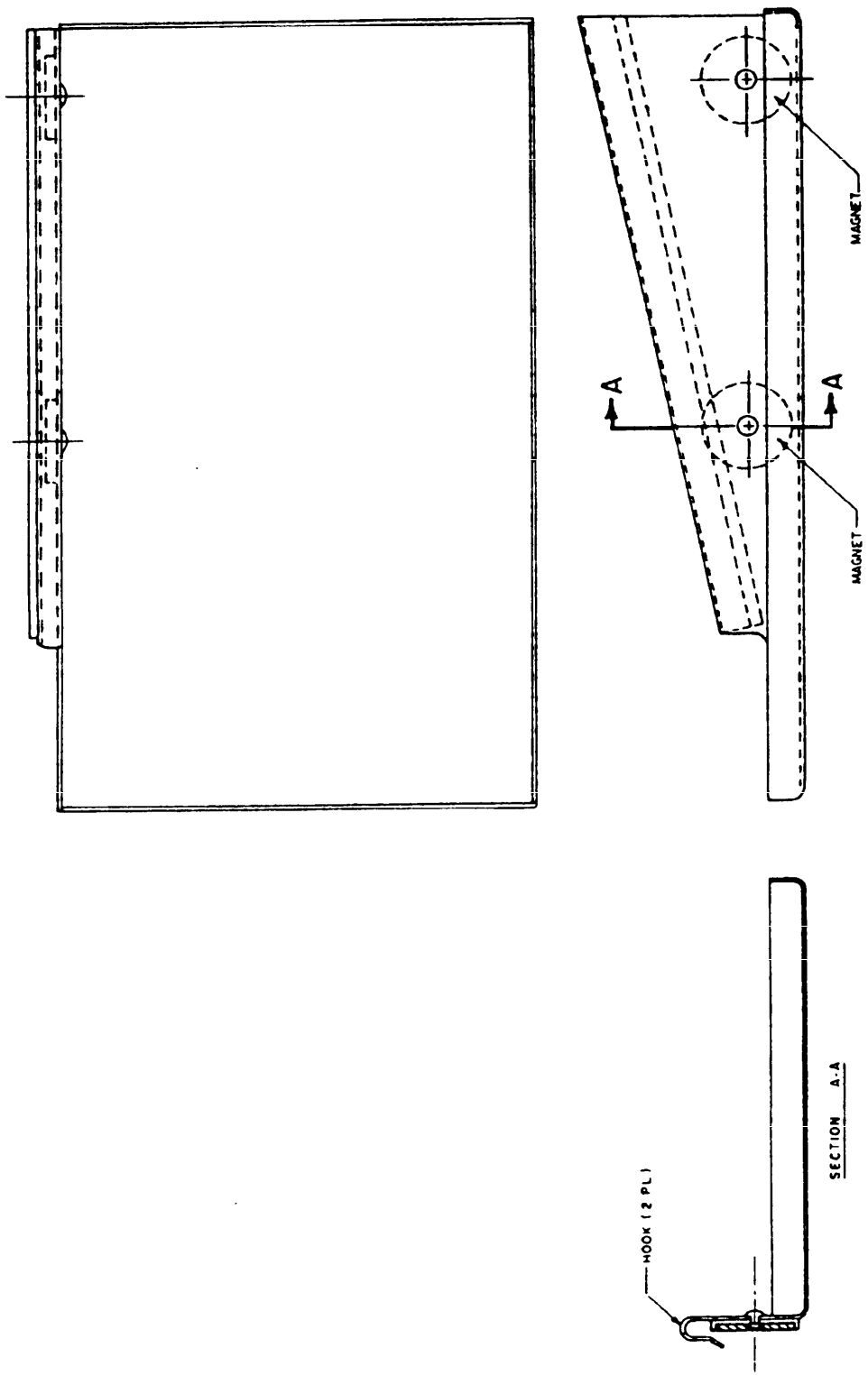
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FIGURE 1. Table, scribing, tracing, drafting.

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FIGURE 2 Tray.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-52991A(ME)	2. DOCUMENT DATE (YYMMDD) 92/08/18

3. DOCUMENT TITLE
Table, Scribing, Tracing, Drafting: Flip Top

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Violet Stewart	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3467 (2) AUTOVON 654-3467
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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